UNIT 2.INTRODUCTION TO DATA MINING

Functionalities of Data Mining.

Classification

It is a data mining technique that groups data into fixed categories based on labeled data.

Example:

A student is classified as **Pass** or **Fail** based on marks, or a loan application is classified as **Approved** or **Rejected**.

Clustering

It groups similar data items together without using predefined labels. The system finds patterns on its own.

Example:

Customers are grouped into different segments (like high spenders, medium spenders, low spenders) based on their purchasing behavior.

Association Rule Mining

It finds relationships between items that frequently occur together in transactions.

Example:

If customers buy **Milk**, they also often buy **Bread** (Milk → Bread).

Prediction / Regression

It is a data mining technique used to estimate or forecast future values by analyzing patterns in past data.

Example: Predicting a student's final exam mark based on their internal assessment scores.

Outlier Analysis

It identifies rare, unusual, or abnormal data that is very different from other data.

Example: Detecting unusual credit card transactions to identify possible fraud.

Summarization

Summarization means reducing a large amount of data into a short and simple form that is easy to understand.

Instead of looking at thousands of data records, we create a small summary that shows the important information.

Example

Consider daily sales data for one week:

Day	Sales (₹)
Monday	2000
Tuesday	2500
Wednesday	1800
Thursday	3000
Friday	2700
Saturday	3500
Sunday	4000

Instead of the above, if it is **summarized** as:

- Total weekly sales = ₹19,500
- Average daily sales = ₹2,785
- Highest sales day = Sunday (₹4,000)
- Lowest sales day = Wednesday (₹1,800)

Now it is easy and quickly understands the performance without checking every day.

Trend and Evolution Analysis

It means **studying data over time to see how it changes**.

This analysis look at past data to find:

- Is it increasing?
- Is it decreasing?
- Does it repeat in a pattern (seasonal)?

Example 1: Mobile Sales

Suppose a shop records mobile sales for 5 years:

Year	Sales
2021	1000
2022	1500
2023	2000
2024	2500

Year	Sales
2025	3000

By observing sales are **increasing every year**.

This increasing pattern is called a **trend**.

Example 2: Ice Cream Sales

Month	Sales
January	200
April	500
May	800
December	150

By observation:

- Sales are **high in summer**
- Sales are **low in winter**

This repeating seasonal pattern is called **evolution or seasonal trend**.

Data Preprocessing

Data preprocessing is the process of cleaning and preparing raw data before applying data mining or machine learning algorithms.

Real-world data is usually:

- Incomplete
- Noisy
- Inconsistent
- Unstructured

So, we preprocess it to improve **data quality** and **model accuracy**.

Why Data Preprocessing is Important?

- Improves accuracy of models
- Reduces errors
- Saves time in analysis
- Makes data suitable for algorithms

Data preprocessing techniques are mainly divided into **5 major categories**:

- Data Cleaning
- Data Integration
- Data Transformation
- Data Reduction
- Data Discretization

Data Cleaning

Removes errors and missing values from data.

Problems handled:

- Missing values
- Duplicate records
- Incorrect data
- Noise

Methods:

- Fill missing values (mean, median, mode)
- Remove duplicates
- Correct inconsistent data

Example Dataset

ID	Name	Age	Gender	Salary
1	Ravi	25	Male	25000
2	Meena		Female	30000

ID	Name	Age	Gender	Salary
3	Ravi	25	Male	25000
4	Kavya	-5	Female	40000
5	John	200	Male	50000

Missing Values

Example:

ID	Name	Age
2	Meena	

Age is missing.

Handling Missing Value

Ages available: **25, 25, -5, 200**

- **Mean** = (25 + 25 + (-5) + 200) / 4 = **61.25**
- **Median** = (25 + 25) / 2 = **25**
- **Mode** = **25**

Best replacement → **Mode = 25** (most reasonable)

Duplicate Records

Example:

ID	Name	Age	Salary
1	Ravi	25	25000
3	Ravi	25	25000

Record 1 and 3 are **duplicate records**.

Solution:

- Remove one duplicate row.

Incorrect Data / Outlier

Example:

Name	Age
Kavya	-5

Age **cannot be negative → Incorrect data**.

Solution:

- Correct using valid range (e.g., Age ≥ 0)
- Replace with mean/median/mode

Noise

Example:

Name	Age
John	200

Age = 200 is **noise**

Solution:

- Remove outlier
- Smooth using binning
- Replace with median

Final Cleaned Dataset (After Preprocessing)

ID	Name	Age	Gender	Salary
1	Ravi	25	Male	25000
2	Meena	25	Female	30000
4	Kavya	25	Female	40000
5	John	25	Male	50000

Data Integration in Data Mining

Data Integration is the process of combining data from multiple sources into a single, consistent, and unified dataset for analysis.

In real-world applications, data is stored in:

- Different databases
- Different file formats (Excel, CSV, SQL)
- Different departments
- Different systems

Before applying data mining techniques, we must integrate all this data.

Why Data Integration is Needed?

When data comes from different sources:

- Column names may be different
- Data formats may vary
- Units may be different
- Duplicate records may exist
- Some values may conflict

Example (Company Sales System)

The data is stored in three different departments.

Source 1: Customer Details (CRM System)

Cust_ID	Name	City
1001	Ravi	Chennai
1002	Meena	Madurai
1003	Arun	Coimbatore

Source 2: Sales Data (Billing System)

CustomerID	Product	Amount
1001	Laptop	50000
1002	Mobile	20000
1003	Tablet	15000

Source 3: Payment Records (Accounts System)

ID	Payment_Mode	Amount_Paid
1001	UPI	50000
1002	Credit Card	20000
1003	Cash	15000

Problems Before Integration

Column names are different:

- Cust_ID
- CustomerID
- ID

Data stored in different systems

No single table shows complete customer information

Step-by-Step Data Integration Process

Step 1: Schema Integration

Rename columns to common name:

Cust_ID, CustomerID, ID → Customer_ID

Step 2: Combine Tables (Using Join Operation)

We combine all three tables using Customer_ID.

Final Integrated Dataset

Customer_ID	Name	City	Product	Amount	Payment_Mode
1001	Ravi	Chennai	Laptop	50000	UPI
1002	Meena	Madurai	Mobile	20000	Credit Card
1003	Arun	Coimbatore	Tablet	15000	Cash

Types of Issues in Data Integration

Schema Conflict

Different column names or structures.
Example:

- DOB in one table
- Date_of_Birth in another

Solution: Rename to common format.

Data Value Conflict

Example:

- Price stored as ₹50000
- Price stored as 50000

Solution: Remove currency symbol and standardize.

Data Redundancy

Same customer appears multiple times.
Solution: Remove duplicates.

Unit Mismatch

Example:
Height in one file = cm
Height in another file = meters
Solution: Convert to same unit.

Data Transformation in Data Mining

Definition:
Data Transformation is the process of converting raw data into a suitable format for analysis or mining so that patterns and insights can be discovered more effectively.

Common Data Transformation Techniques

Normalization / Scaling

- Converts numeric values into a common scale.
- **Example:** Age 16–40 → scaled to 0–1
- Useful for distance-based algorithms (like K-Means, KNN).

Smoothing (Binning)

- Reduces noise in numeric data by **grouping into bins**.
- **Example:** Ages 16, 18, 20, 21, 22 → Bin 15–25 → replace with mean 19
- Makes trends easier to detect.

Aggregation

- Combine multiple records into summary values.
- **Example:** Daily sales → monthly sales total.
- Reduces data size and highlights patterns.

Generalization

- Replace detailed data with higher-level concepts.
- **Example:** Age 16, 18, 20 → Young; 35, 40 → Adult
- Often uses concept hierarchies.

Attribute / Feature Construction

- Create new features from existing ones.
- **Example:** Profit = Revenue – Cost

- Improves model performance.

Encoding Categorical Data

- Convert text categories to numeric values for algorithms.
- **Example:** Mobile Type: iPhone → 1, Touch Phone → 2, Button Phone → 3

Discretization

- Convert continuous values into categories.
- **Example:** Age 16–25 → Young, 26–40 → Adult, 41+ → Elder

Binning in Data Mining

Definition:
Binning is a data preprocessing technique where continuous numerical data is divided into intervals (bins) and each value in a bin is replaced by a representative (mean, median, or boundary).

Purpose:

- Reduce noise and smooth data
- Make patterns easier to detect

Example Data

Age : 20,21,25,16,18,28,30,32,35,40

Step 1: Sort Data

16, 18, 20, 21, 22, 25, 30, 32, 35, 40

Step 2: Decide Number of Bins

Let's create 3 bins.

Step 3: Create Bins (Equal-Width Method)

- Min = 16, Max = 40 → Range = min – max = 24
- Bin width = 24 ÷ 3 = 8

Bins:

1. Bin 1 → 16–24
2. Bin 2 → 25–32
3. Bin 3 → 33–40

Step 4: Replace Values with Bin Representative

- Bin 1 (16–24): 16,18,20,21,22 → Mean = 19
- Bin 2 (25–32): 25,30,32 → Mean = 29
- Bin 3 (33–40): 35,40 → Mean = 37.5 → 38

Binned / Smoothed Dataset:

Record	Age (Original)	Age (Binned)
1	16	19
2	18	19
3	20	19
4	21	19
5	22	19
6	25	29
7	30	29
8	32	29
9	35	38
10	40	38

Data Reduction

Data Reduction is a preprocessing technique used to reduce the size of data while maintaining its important information.

Why Data Reduction is Needed?

Large datasets:

- Take more memory

- Increase computation time
- Slow down machine learning models

Data reduction helps in:

- Faster model training
- Better performance
- Reduced storage cost

Techniques of Data Reduction

Data Cube Aggregation

Combine detailed data into summarized data at a higher level.

Example: Supermarket Sales Data

Original Data (Daily Sales)

Date	Product	City	Sales
Jan 1	Milk	Delhi	100
Jan 2	Milk	Delhi	120
Jan 3	Milk	Delhi	130
Jan 1	Bread	Delhi	80
Jan 2	Bread	Delhi	90

After Aggregation (Monthly Sales)

Month	Product	City	Total Sales
January	Milk	Delhi	350
January	Bread	Delhi	170

Instead of storing daily details (many rows), store monthly totals (few rows).

✓ Number of records reduced

✓ Still useful for decision-making

Dimensionality Reduction

Reducing the number of features (columns) in a dataset while keeping important information.

Common Methods:

- Feature selection
- Principal Component Analysis (PCA)

Example Dataset: Student Marks

Suppose the dataset has **5 students** and their marks in **5 subjects**:

Student	Math	Science	English	Social	Computer
A	85	88	78	90	92
B	70	75	72	68	74
C	90	92	85	88	95
D	60	65	70	62	68
E	80	82	75	85	88

Step 1: Identify Correlations

- Students who score high in Math often score high in Science and Computer → correlated features.

Step 2: Remove Irrelevant Features (Feature Selection)

Suppose we decide **English and Social** are less important for overall performance prediction.

Reduced dataset:

Student	Math	Science	Computer
A	85	88	92
B	70	75	74
C	90	92	95
D	60	65	68
E	80	82	88

- **Before:** 5 features → **After:** 3 features
- Still preserves most information about academic ability.

Step 3: Combine Features (Feature Extraction / PCA)

combine **Math + Science + Computer** into a single new feature called **Academic Score**.

Student	Academic Score
A	(85+88+92)/3 = 88.3
B	(70+75+74)/3 = 73.0
C	(90+92+95)/3 = 92.3
D	(60+65+68)/3 = 64.3
E	(80+82+88)/3 = 83.3

Now, **5 dimensions → 1 dimension** (Academic Score).

Data Compression

Data Compression is a data reduction technique used to reduce the size of data while keeping useful information for analysis.

There are **two types**:

1. Lossless Compression
2. Lossy Compression

Lossless Compression

Original data can be fully recovered

Name	Mark
Ravi	80
Ravi	80
Meena	75
Meena	75
Meena	75

Here, records are repeated.

After Lossless Compression

Name	Mark	Count
Ravi	80	2
Meena	75	3

- Instead of storing 5 rows, we store 2 rows + count.
- No information is lost.

<u>Lossy Compression</u>

Some original information is lost

Original Data

Name	Mark
Ravi	81
Meena	79
Arjun	82
Divya	80

Suppose:

- 80–100 → A
- 70–79 → B

Name	Grade
Ravi	A
Meena	B
Arjun	A
Divya	A

- ❖ Exact marks (81, 79, 82, 80) are removed.
- ❖ Only Grade is stored.
- ❖ We cannot get original marks back.

This is **Lossy Compression**.

Numerosity Reduction

Replaces original data with a smaller representation.
Methods:

- Clustering

Example:
Instead of storing all data points, store cluster centers.

Data Discretization

Data Discretization is the process of converting **continuous numerical values into categories (groups or ranges).**

Common Methods of Data Discretization

- ❖ Equal Width Binning
- ❖ Equal Frequency Binning
- ❖ Clustering-based

Example Dataset (Marks of 10 Students)

Marks:

10, 20, 30, 40, 50, 60, 70, 80, 90, 100

Equal Width Binning

Divide the data range into **equal-sized intervals**.

Step 1: Find range

Min = 10
Max = 100
Range = 100 − 10 = 90

Bins = 3

Bin width = 90 / 3 = 30

Step 2: Create bins

- 10–40
- 41–70
- 71–100

Step 3: Assign values

Marks	**Bin**
10	10–40
20	10–40
30	10–40
40	10–40
50	41–70
60	41–70
70	41–70
80	71–100
90	71–100
100	71–100

Equal Frequency Binning

Each bin contains **equal number of data points**.

10 values presetn.
2 bins → each bin gets 5 values.

Step 1: Sort data

10, 20, 30, 40, 50, 60, 70, 80, 90, 100

Step 2: Divide into 2 equal groups

Bin 1: 10, 20, 30, 40, 50
Bin 2: 60, 70, 80, 90, 100

Now convert into ranges:

- 10–50
- 60–100

Each bin has **equal number of values (5)**
But range is not equal.

Clustering-Based Discretization

Group the values based on similarity (like k-means clustering).

Suppose we create 2 clusters:

Cluster 1 (Low marks):
10, 20, 30, 40

Cluster 2 (High marks):
50, 60, 70, 80, 90, 100

Now we label:

- Low
- High

So:

Marks	Category
10	Low
90	High

Concept Hierarchy (Multiple Levels)

It organizes data into **multiple levels of abstraction**.

Example:

Marks → Grade → Performance

Marks	Grade	Performance
92	A	Excellent
75	B	Good
40	D	Poor

Data Mining

Data mining is the process of extracting useful patterns and knowledge from large amounts of data to support decision making.

A **pattern** is a **useful relationship or repeated behavior** found in data.

Simple Examples

Supermarket Example

Data:

- Milk, Bread, Butter
- Milk, Bread
- Milk, Bread, Jam

Pattern found:

Customers who buy **Milk** often buy **Bread**

Education Example

Data:

- Low attendance → low marks
- Low attendance → low marks

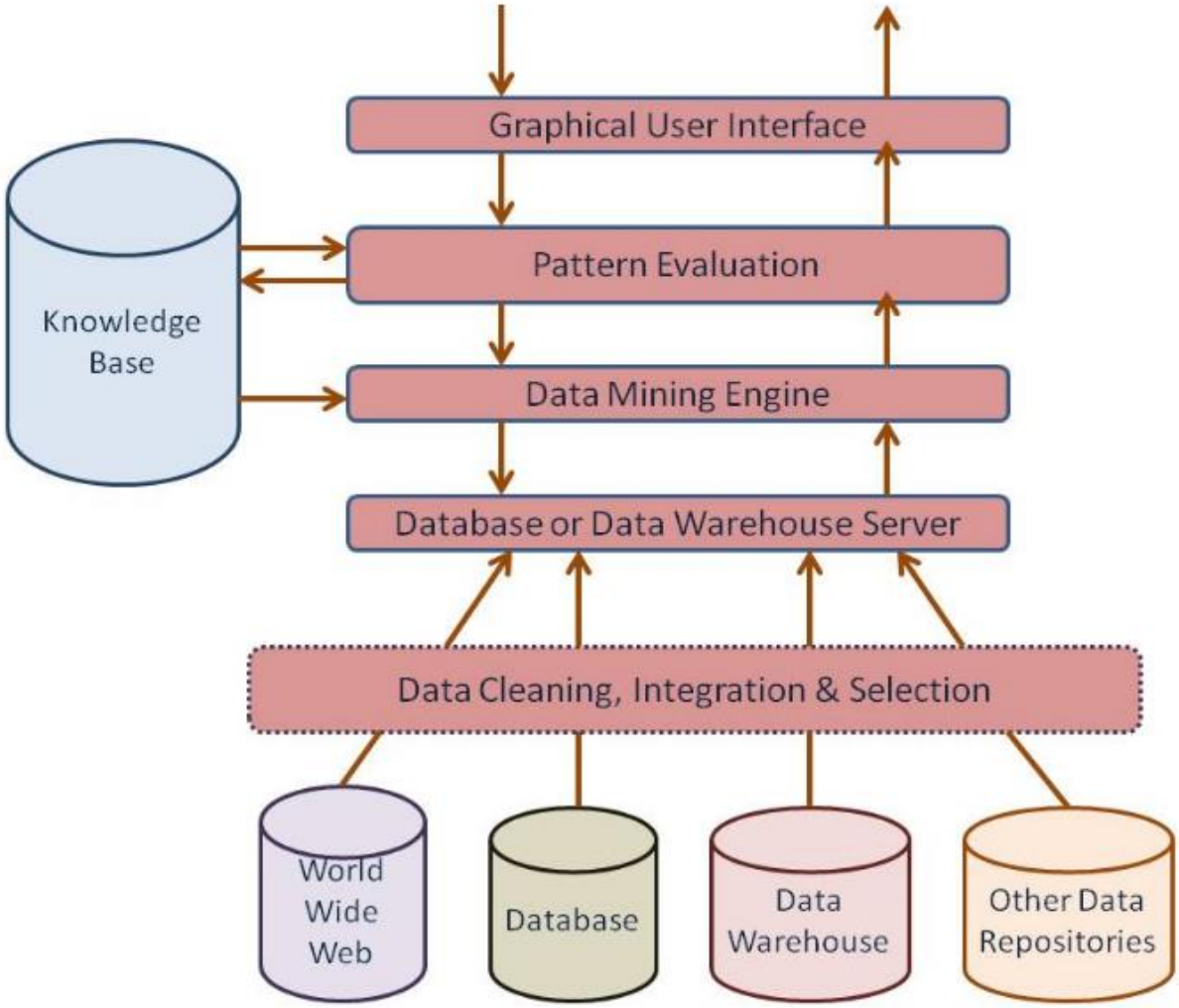
Pattern found:

Students with **low attendance** usually get **low marks**

Why do we need Data Mining?

- Huge amounts of data are generated every day and stored
- Manually analyzing data is difficult
- Data mining helps us:
    - Understand data
    - Find hidden patterns
    - Make better decisions

Data Mining Architecture



Data Sources

These are the places where data comes from:

- **Database** – operational data (student records, bank transactions)
- **Data Warehouse** – integrated historical data
- **World Wide Web** – web pages, logs, social media data
- **Other Information Repositories** – files, documents, spreadsheets

This data is usually **raw**, **large**, and **unprocessed**.

Data Cleaning, Integration and Selection

This is a **preprocessing stage**.

Data Cleaning

- Removes missing values
- Removes noise and duplicate data

  Noise = wrong or irrelevant data (Example: Mark = -8)

Data Integration

- Combines data from multiple sources

Data Selection

- Selects only the required data for mining

Output: **Clean, relevant data** ready for mining

Database or Data Warehouse Server

- Stores the **cleaned and integrated data**
- Handles:
  - Data retrieval
  - Query processing
- Supplies data to the **Data Mining Engine**

Acts as a **bridge** between raw data and mining algorithms

Data Mining Engine

This is the **core component** of the system

- It Applies data mining algorithms such as:
    - Classification
    - Clustering
    - Association rule mining
    - Prediction
    - Outlier detection

Extracts **patterns and knowledge** from data

Pattern Evaluation

Pattern evaluation checks the patterns found by the data mining process. It selects only **interesting, useful, and meaningful patterns** and removes **irrelevant or repeated patterns**, so that only valuable knowledge is kept.

Example:

In a supermarket database, many rules are found. The rule **"Customers who buy milk also buy bread"** is kept because it occurs frequently, while a rule like **"Customers who buy milk also buy pen"** is removed because it is rare and not useful.

Knowledge Base

The knowledge base stores important information like rules and patterns found from data. It helps the system remember past knowledge and use it again when analyzing new data.

Example:

If the system already knows the rule **"students with low attendance may fail,"** it uses this rule to check and predict results for new students.

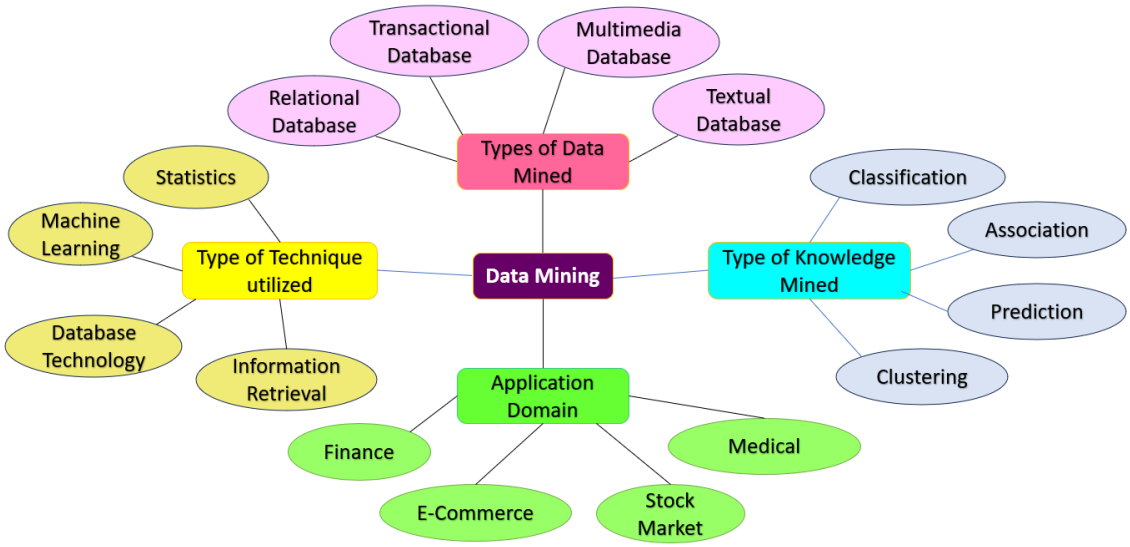
- ❖ Acts like the **memory** of the data mining system

User Interface

- The user interface is the part of the system that allows users to interact with data mining. Users can give inputs and see the results clearly.
- Displays results in:
  - Charts
  - Tables
  - Graphs
  - Reports

❖ Makes the system **interactive and user-friendly**

Classification of Data Mining Systems



1. Based on the Type of Data Mined

Relational Data Mining

Works on structured data stored in tables, where information is organized into rows and columns in a database.

For example, student marks stored in a database table can be easily analyzed and managed.

Transactional Data Mining

Transactional Data Mining deals with transaction records that contain sets of items purchased or occurring together.

For example, supermarket purchase transactions are analyzed to find frequently bought item combinations

Data Warehouse Mining

Data Warehouse Mining mines large volumes of historical data collected from multiple sources and stored in a centralized repository.

For example, year-wise company sales data can be analyzed to identify growth trends and performance patterns.

Spatial Data Mining

Spatial Data Mining studies data related to places and locations. It finds patterns based on area, distance, and position.

For example, by looking at accident data on a map, we can find accident-prone areas and improve road safety in those locations.

Temporal / Time-Series Data Mining

It analyzes data that is collected again and again over time (daily, monthly, yearly) to find patterns or trends.

Examples:

- Daily temperature recorded for one year to see weather changes.
- Monthly electricity bill to see usage increase or decrease.
- Daily stock price to see whether price is going up or down over time.

Text and Multimedia Mining

It extracts useful information from text, images, audio, and video files. This type of data is usually unstructured (not in table format).

Examples:

- Analyzing customer reviews (text) to find positive or negative opinions.
- Detecting disease from X-ray or MRI images (image data).
- Converting speech to text from audio recordings.
- Identifying objects or faces in videos.

2. Based on the Type of Knowledge Discovered

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3. Based on Techniques Used

Statistical Methods

Uses math formulas to analyze data.

Example: Finding class average marks or highest and lowest marks.

Machine Learning Methods

Computer learns from past data and makes decisions.

Example: Predicting whether a student will pass or fail.

Neural Networks

A smart system that works similar to the human brain.

Example: Recognizing a person's face in a photo.

Graph-Based Mining

Shows data as points and connections between them.

Example: Finding how people are connected in a social network.

4. Based on Application Area

Business and Marketing

Data mining helps businesses understand customer buying behavior and make better decisions.

Example:

Recommending products to customers based on what they bought earlier.

Banking and Finance
Data mining is used to check risk and find fraud in financial activities.
Example:
Deciding whether to approve or reject a loan.

Healthcare
Data mining helps doctors analyze patient data to identify diseases early.
Example:
Predicting heart disease using medical test data.

Education
Data mining analyzes student data to improve academic performance.
Example:
Predicting student results using marks and attendance.